

Appendix C - Table C.1: Custom PCB Bill of Materials										
Installed PCB, components, modules, protection devices, connectors, and IC sockets.										
All quantities correspond to one completed system. Purchased spares and unused multipack quantities are excluded.										

Item	Cost category	Designators	Qty.	Component / value	Key specification	Manufacturer	Manufacturer part number	Allocated unit cost	Extended cost	Source / allocation note
1	Custom PCB fabrication	PCB-1	1	Custom four-layer PCB	1.6 mm nominal board; one usable board installed	JLPCB	Custom BMS PCB	\$17.4420	\$17.44	JLPCB order total allocated across five boards
2	Control and sensing	Op-amp IC (Altium export: 1)	1	MCP6002-I/P	Dual general-purpose operational amplifier, 8-pin PDIP	Microchip Technology	MCP6002-I/P	\$0.5950	\$0.60	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
3	Interconnect and protection	AUX, LED, MENU	3	2-position JST-XH headers	Vertical, 2.50 mm pitch	JST	B2B-XH-A(LF)(SN)			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
4	Passives	C1	1	2.2 nF ceramic capacitor	250 V, COG, 5%, radial	KEMET	C325C222JAG5TA	\$0.7708	\$0.77	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
5	Passives	C2, C3, C5, C7, C8, C10, C11, C13, C14, C18, C19, Cbot, Cdriver, CIN2	14	0.1 µF ceramic capacitors	50 V, X7R, radial	Vishay	K104K15X7RF5TL2	\$0.2191	\$3.07	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
6	Passives	C4, C15, C16, C17, C20	5	10 nF ceramic capacitors	25 V, COG, 10%, radial	KEMET	C320C103K3G5TA	\$0.4666	\$2.33	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
7	Passives	C6	1	470 µF aluminum electrolytic capacitor	35 V, radial	Rubycon	35PX470MEFC10X12.5	\$0.8384	\$0.84	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
8	Passives	C9, C12	2	10 µF ceramic capacitors	25 V, X7R, radial	TDK	FG26X7R1E106KRT06	\$1.0413	\$2.08	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
9	Passives	CIN1, COUT1	2	220 µF aluminum electrolytic capacitors	35 V, low-impedance radial	Rubycon	35ZLH220MEFC8X11.5	\$0.5815	\$1.16	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
10	Passives	COUT2	1	22 µF ceramic capacitor	25 V, X7S, radial	Murata Electronics	RDEC71E226K3P1H03B	\$1.9744	\$1.97	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
11	Power conversion	D1	1	STP55L60 Schottky rectifier	60 V, 5 A, DO-201AD	STMicroelectronics	STP55L60	\$1.1495	\$1.15	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
12	Power conversion	D2	1	1N5819-T Schottky diode	40 V, 1 A, DO-41; bootstrap diode	Diodes Incorporated	1N5819-T	\$0.7438	\$0.74	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
13	Interconnect and protection	D3	1	1.5KE16A TVS diode	Unidirectional, 13.6 V standoff, 1500 W	Littelfuse	1.5KE16A	\$1.4335	\$1.43	Purchased part; Altium value was listed as 1.5KE16AG. DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
14	Control and sensing	D4	1	1N4148 switching diode	100 V, 200 mA, DO-35	onsemi	1N4148	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
15	Interconnect and protection	ENABLE, OLED	2	4-position JST-XH headers	Vertical, 2.50 mm pitch	JST	B4B-XH-AM(LF)(SN)			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
16	Interconnect and protection	F1 holder	1	PCB blade-fuse holder	ATO/ATC style, through-hole	Keystone Electronics	3568	\$1.5822	\$1.58	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
17	Interconnect and protection	F1 fuse	1	5 A automotive blade fuse	32 VDC, ATO/ATC	Littelfuse	0287005.PXCN	\$0.4868	\$0.49	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
18	Control and sensing	IC1	1	3.3 V DC-DC regulator	5 W, 1.5 A, through-hole switching regulator	Murata Power Solutions	OKI-78SR-3.3/1.5-W36-C	\$7.6542	\$7.65	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
19	Interconnect and protection	J1	1	PCB-mount XT60 connector	Right-angle male battery connector	AMASS	XT60PW-M			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
20	Interconnect and protection	J2	1	2-position output terminal block	6.35 mm pitch, through-hole	Phoenix Contact	1714955			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
21	Interconnect and protection	J3, J4	2	4-position JST-XH balance headers	Vertical, 2.50 mm pitch	JST	B4B-XH-A	\$0.2299	\$0.46	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
22	Interconnect and protection	J5, J6	2	4-position selector-interface headers	CK1027 pole wiring interfaces, 2.50 mm pitch	JST	B4B-XH-AM(LF)(SN)			These are PCB wiring headers, not two additional rotary switches; cost included in connector set
23	Control and sensing	J_ADC1, J_ADC2	2	ADS1115 ADC modules	16-bit, four-channel I2C ADC breakout modules	HiLetgo	ADS1115 module	\$3.9633	\$7.93	Two modules used from a three-module purchase; only the installed share is allocated
24	Interconnect and protection	J_PROG	1	4-pin programming header	2.54 mm pitch, vertical	Würth Elektronik	61300411121			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
25	Control and sensing	K1, K2	2	TQ2-3V DPDT relays	3 V coil, 2 Form C	Panasonic Industry	TQ2-3V	\$5.0725	\$10.15	DigiKey invoice 129154402; two of four relays allocated to this unit
26	Power conversion	L1	1	68 µH power inductor	2.9 A rated, 80 mΩ, unshielded, through-hole	Würth Elektronik	7447480680	\$2.2449	\$2.24	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax

27	Power conversion	Q1	1	N-channel power MOSFET	60 V, 37.2 A, 12 mΩ, TO-220AB	Diodes Incorporated	DMT6009LCT	\$2.0555	\$2.06	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
28	Control and sensing	Q2, Q3	2	2N3904 NPN transistors	40 V, 200 mA, TO-92	onsemi	2N3904BU	\$0.3787	\$0.76	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
29	Control and sensing	Q4	1	2N2222A NPN transistor	50 V, 800 mA, TO-18	Microchip Technology	2N2222A	\$3.9218	\$3.92	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
30	Passives	R1, R2, R3, R4, R6, R16, R17, R19, R27, R38, R44, R_BOT	12	10 kΩ resistors	1%, 1/4 W, axial	Yageo	MFR-25FRF52-10K	\$0.0568	\$0.68	Twelve installed; invoice unit price used as the allocation basis. DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
31	Passives	R5	1	3.32 kΩ resistor	1%, 1/4 W, axial	Stackpole Electronics	RNF14FTD3K32	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
32	Passives	R7	1	26.1 kΩ resistor	1%, 1/4 W, axial	Yageo	MFR-25FRF52-26K1	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
33	Passives	R8	1	22 Ω resistor	1%, 0.6 W, axial	Yageo	MF0207FTE52-22R	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
34	Passives	R9, R10, R11, R20, R28, R32, R33, R35	8	100 kΩ resistors	1%, 1/4 W, axial	Stackpole Electronics	RNF14FTD100K	\$0.0498	\$0.40	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
35	Passives	R12	1	0.05 Ω current-shunt resistor	1%, 3 W, axial	Stackpole Electronics	MR3FT50L0	\$2.6100	\$2.61	Correct purchased shunt; raw Altium description referenced a different resistor. DigiKey invoice 128543474; unit price with proportional tariff
36	Passives	R13	1	51 kΩ resistor	1%, 1/4 W, axial	Yageo	MFR-25FRF52-51K	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
37	Passives	R14	1	191 kΩ resistor	1%, 1/4 W, axial	Yageo	MFR-25FRF52-191K	\$0.1352	\$0.14	Purchased value used; Altium rounded the value to 190 kΩ. DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
38	Passives	R15	1	30 kΩ resistor	1%, 0.6 W, axial	Yageo	MF0207FTE52-30K	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
39	Passives	R18, R41	2	47 kΩ resistors	1%, axial	Yageo; TE Connectivity	MFR-12FTF52-47K; LR1F47K	\$0.1352	\$0.27	Two installed parts at the same invoice unit price. DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
40	Passives	R21, R24, R25, R26	4	330 kΩ resistors	1%, 0.6 W, axial	TE Connectivity	LR1F330K	\$0.0554	\$0.22	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
41	Passives	R22	1	680 kΩ resistor	1%, 1/2 W, axial	Yageo	MFR50SFE52-680K	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
42	Passives	R23	1	1.3 MΩ resistor	1%, 1/4 W, axial	Yageo	MFR-25FBF52-1M3	\$0.1488	\$0.15	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
43	Passives	R29, R30, R31, R37, R43, R45	6	1 kΩ resistors	1%, 1/4 W, axial	Yageo	MFR-25FRF52-1K	\$0.0568	\$0.34	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
44	Passives	R34	1	470 Ω resistor	1%, 1/4 W, axial	Yageo	MFR-25FRF52-470R	\$0.1352	\$0.14	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
45	Control and sensing	R36	1	10 kΩ NTC thermistor	B = 3380 K, radial bead	Murata Electronics	NXRT15XH103FA1B040	\$0.4733	\$0.47	Correct purchased thermistor; raw Altium description listed 3950 K. DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
46	Passives	R39, R40, R42, R46	4	220 Ω resistors	Axial through-hole; used for LED/current limiting	Generic / existing stock	Not documented	\$0.1352	\$0.54	Installed parts are included, but no purchase record or unit price was provided
47	Interconnect and protection	S2	1	6-position JST-XH header	Vertical, 2.50 mm pitch; mode-switch interface	JST	B6B-XH-AM(LF)(SN)			Cost included in the \$1.50 miscellaneous connector set on Off-board BOM
48	Power conversion	U1	1	TL494 PWM controller	16-pin PDIP	Texas Instruments	TL494CN	\$0.7708	\$0.77	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
49	Power conversion	U2	1	High-side/low-side gate driver	500 V half-bridge driver, 14-pin PDIP	Infineon Technologies	IRS2110PBF	\$6.1666	\$6.17	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
50	Control and sensing	U3	1	MSP430 microcontroller	16 MHz, 16 KB Flash, 20-pin PDIP	Texas Instruments	MSP430G2553IN20	\$5.5310	\$5.53	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
51	Control and sensing	U4	1	Battery current sensor	Unidirectional Hall-effect sensor, 50 A	Allegro MicroSystems	ACS758LCB-050U-PFF-T	\$14.6187	\$14.62	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
52	Control and sensing	U5	1	Buck-output current sensor	GMR current sensor, 2.5 A range, SOIC-8	Allegro MicroSystems	ACS70331EOLCTR-2P5U3	\$3.8812	\$3.88	DigiKey invoice 128543474; unit price with proportional tariff, shipping, and tax
53	IC sockets	U1 socket	1	16-pin machine-pin IC socket	0.3-inch DIP socket	Adam Tech	ICM-316-1-GT-HT	\$2.4257	\$2.43	DigiKey invoice 128551076; allocated delivered unit
54	IC sockets	U2 socket	1	14-pin machine-pin IC socket	0.3-inch DIP socket	Adam Tech	ICM-314-1-GT-HT	\$2.8073	\$2.81	DigiKey invoice 128551076; allocated delivered unit
55	IC sockets	Op-amp socket	1	8-pin machine-pin IC socket	0.3-inch DIP socket	Adam Tech	ICM-308-1-GT-HT	\$1.9624	\$1.96	DigiKey invoice 128551076; allocated delivered unit
56	IC sockets	U3 socket	1	20-pin IC socket	0.3-inch DIP socket	CNC Tech	245-20-1-03	\$2.5347	\$2.53	DigiKey invoice 128551076; allocated delivered unit

Appendix C - Table C.2: Off-board System Bill of Materials

Installed battery, charger, display, controls, indicators, enclosure, wiring, and hardware.

All quantities correspond to one completed system. Purchased spares and unused multipack quantities are excluded.

Item	Cost category	Designators	Qty.	Component / value	Key specification	Manufacturer	Manufacturer part number	Allocated unit cost	Extended cost	Source / allocation note
1	Battery and charging	BAT1	1	3S LiPo battery pack	11.1 V nominal, 2200 mAh, 50C; XT60 and JST-XH leads	Zeee	3S 2200 mAh pack	\$10.9950	\$11.00	One battery used from a two-battery set
2	Battery and charging	CHG1	1	2S/3S LiPo balance charger	Internal charger assembly connected to external charging port	SSUPCHG	ZY-B3PRO-12W / 2S-3S XHR charger	\$12.9900	\$12.99	Actual purchase price was \$12.99; current listing price is not used
3	User interface	DISP1	1	2.42-inch OLED module	128 × 64, SSD1309, I2C interface	HiLetgo	2.42-inch SSD1309 module	\$5.9900	\$5.99	Actual purchase price was \$5.99; current listing price is not used
4	User interface	SW_SEL	1	Three-position rotary selector	Four-pole, three-position rotary switch	Lorlin	CK1027	\$11.6700	\$11.67	Mouser order 91197944; cost for the one switch
5	User interface	KNOB1	1	Aluminum selector knob	1/4-inch shaft, silver, indicator mark	KAISH	Universal aluminum control knob	\$4.9950	\$5.00	One knob used from a two-knob package; only the installed share is
6	User interface	SW_MODE	1	ON-OFF-ON operating-mode toggle switch	DPDT, six-terminal, panel-mount	Nilight	6-pin heavy-duty toggle switch	\$1.9780	\$1.98	One switch used from a five-switch package;
7	User interface	PB1	1	Illuminated momentary	16 mm, 1NO, blue 3 V LED, IP67	Generic	16 mm illuminated pushbutton	\$2.9900	\$2.99	Actual purchase price was \$2.99
8	User interface	LED_3V3, LED_5V, LED_9V	3	Voltage-selection indicator LEDs	5 mm, 20 mA, clear lens	Generic	5 mm LED assortment	\$0.1299	\$0.39	Three LEDs installed; the remaining LEDs and their cost are excluded from
9	Enclosure and hardware	ENC1	1	Custom 3D-printed enclosure	Project-specific case, panel, and integrated wire-management features	Custom fabricated	3D-printed enclosure	\$12.4600	\$12.46	Actual filament consumed for the completed enclosure
10	Wiring and connections	CONN_SET	1	Miscellaneous connector set	Non-DigiKey headers/plugs and panel charging connector used in the completed	Various	Project connector set	\$1.5000	\$1.50	all other connectors
11	Wiring and connections	WIRING	1	Internal and output wiring	18 AWG buck-output leads plus low-current signal/charging wiring	Generic / existing stock	Not documented			Installed wiring is listed, but no material cost
12	Enclosure and hardware	HW_SET	1	Mounting hardware	PCB standoffs, screws, nuts, and associated enclosure hardware	Generic / existing stock	Not documented			Existing inventory

Appendix C - Cost Summary for One Completed System

Costs are allocated only to the quantities installed. Spare parts and unused multipack quantities are excluded.

Summary metric	Value
PCB BOM cost	\$118.43
off-board BOM cost	\$65.96
cost of one completed system	\$184.39

Cost category	Allocated cost
Custom PCB fabrication	\$17.44
Control and sensing	\$55.64
Power conversion	\$13.13
Passives	\$18.52
Interconnect and protection	\$3.96
IC sockets	\$9.73
Battery and charging	\$23.99
User interface	\$28.01
Enclosure and hardware	\$12.46
Wiring and connections	\$1.50
Total	\$184.39

Allocation basis	Source amount	Allocation divisor / factor
DigiKey main invoice landed-cost factor	\$117.45	\$86.85
DigiKey socket invoice landed-cost factor	\$9.73	\$3.57
DigiKey relay invoice landed-cost factor	\$20.29	\$9.84
JLCPCB delivered cost per board	\$87.21	5
Battery cost per installed pack	\$21.99	2
ADS1115 cost per installed module	\$11.89	3
Toggle-switch cost per installed switch	\$9.89	5
LED cost per installed LED	\$12.99	100

Method note

DigiKey unit costs include a proportional allocation of tariff, shipping, and sales tax from each invoice. Other multipack purchases are divided by the number of items in the package, then multiplied only by the quantity installed. No purchased spare, replacement, or unused component is included in the BOM total.